

Yes Because It Knows the Item, or Because It Knows the User? Decomposing the Confidence of LLM Recommenders

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Abstract

Large language models are increasingly used as recommenders that answer “Would this user like this item?” with a *Yes* or *No*, and the probability of *Yes* is used as both a ranking score and a confidence. We ask what that confidence actually measures. **[DATA NEEDED: one-sentence observation result: calibration of token-level P(Yes) against rated labels on 4 datasets $\times$ 2 backbones; share of confident errors; popularity dependence of confidence level and of the confidence-matched calibration gap]. [DATA NEEDED: mechanism sentence: item-level acquiescence term, its popularity link, causal pseudonym evidence].** We propose MIRROR, which asks the mirrored question and scores the difference, cancelling the item-level term while keeping the user-specific evidence, and *mirror tuning*, which preserves the cancellation under fine-tuning. **[DATA NEEDED: main result sentence with UAUC / tail calibration gains vs equal-compute ensembles and item-prior subtraction, and next-item ranking vs 8 official baselines].**

CCS Concepts

• Information systems $\rightarrow$ Recommender systems.

Keywords

LLM-based recommendation, uncertainty, calibration, popularity bias

1 Introduction

Recommenders built on large language models (LLMs) routinely reduce recommendation to a binary judgment—“Would this user like this item? Answer Yes or No”—and read the probability of *Yes* as the recommendation score [4, 38]. The same number is then implicitly treated as the model’s *confidence*: a high P(Yes) is taken to mean that the model is sure the user will like the item. Whether that reading is justified has, to our knowledge, never been tested against labels that ask the same question. Prior work quantifies the uncertainty of list-wise LLM recommenders [19], shapes training rewards with confidence [9], or audits verbalized confidence of zero-shot models against catalog membership [30]; none asks *what the yes/no confidence of a personalised LLM recommender is made of*.

We study six questions that practitioners ask of such a system: (Q1) when the model is right, is it sure? (Q2) when it is wrong, is that because it was unsure—or is it confidently wrong? (Q3) does acting on high confidence over-expose popular items? (Q4) how well does the yes/no confidence track ground truth? (Q5) are popular items systematically more confidently endorsed than niche ones, and is that endorsement warranted? (Q6) can confidence identify noisy training data?

Key observation. **[DATA NEEDED: Pilot-1/B3 result: a(i) spread, popularity link, share of confident errors explained by the item term; pseudonym knockout].** Our central hypothesis is that P(Yes) decomposes into an *item term*—a willingness to say *Yes* to an item regardless of who asks, which tracks how familiar or popular the item is—and a *personal-evidence term*. Only the latter is a recommendation signal. Crucially, the item term is not removed by within-user calibration: any monotone recalibration of P(Yes) leaves a user’s ranking unchanged [21], whereas an item-dependent offset reorders it.

Method. MIRROR asks the mirrored question (“Would this user *dislike* this item?”) and scores the difference of the two log-odds. The item term enters both answers with the same sign and cancels; user-specific evidence enters with opposite signs and is doubled. The two-prompt symmetrization estimator itself is not new [6]; our contribution is to show that in *personalised* recommendation the artifact it removes is item-specific and popularity-graded, that removing it changes rankings in a way that equal-compute prompt ensembles and item-prior subtraction do not [16, 40], and that mirror tuning keeps the benefit after fine-tuning.

Contributions. (1) The first calibration study of token-level yes/no confidence of LLM recommenders against rated labels that match the question, answering Q1–Q5 on **[DATA NEEDED: 4 datasets $\times$ 2 backbones]**, with a validity check against fully observed feedback [13]. (2) Evidence that the confidence contains an item-level, popularity-graded acquiescence term, including a causal pseudonym-knockout test with a real-brand placebo. (3) MIRROR and mirror tuning, with **[DATA NEEDED: main gains]**. (4) A characterised set of what does *not* work (within-user recalibration, uncertainty-weighted pruning), reported with multi-seed controls (Appendix).

2 Related Work

LLM-based recommendation. LLMs are used as recommenders in two main forms. *Discriminative* recommenders pose a binary question about a user–item pair and score the item by the probability of answering *Yes*; TALLRec [4] instruction-tunes an LLM on such questions, CoLLM [38] and BinLLM [36] inject collaborative signals into the same yes/no formulation, and LLaRA [25] aligns sequential representations with the language space. These works evaluate P(Yes) only as a discriminative score (AUC or per-user UAUC); none reports its calibration, and none asks what the probability is made of. *Generative* recommenders produce item identifiers or titles directly [2, 14], with decoding corrections [3] and preference-optimisation objectives [8]; SPRec and Flower show that SFT and DPO amplify popularity bias in such models [11, 12]. We build on the discriminative formulation because its output is an explicit probability, and we revisit the same-candidate next-item protocol and baseline set of our earlier controlled study.

Uncertainty in LLM recommenders. UQRec [19] quantifies the predictive uncertainty of list-wise LLM recommendation from a single inference, decomposes it into recommendation and prompt uncertainty, and uses it to choose prompts. UGR [9] trains generative recommenders with an uncertainty-weighted reward that penalises confident errors and uses confidence tokens for request rejection and list truncation. EviRank [35] estimates position-level confidence for LLM ranking lists. Ravikumar [30] audits the verbalized confidence of zero-shot LLM recommenders against catalog membership, stratified by item popularity. KnowSA [22] estimates what an LLM knows about each item and augments prompts only for low-knowledge items, finding popularity a poor proxy for that knowledge. Our work differs in its object and its question: we analyse token-level yes/no confidence of zero-shot *and fine-tuned* personalised recommenders against labels that match the question, decompose it into an item term and a personal-evidence term, and use the decomposition to change rankings.

Confidence and calibration of LLMs. Black-box confidence elicitation shows that verbalized confidence of LLMs is concentrated at high values and only weakly separates correct from incorrect answers [34]; LLMs nevertheless “mostly know what they know” when asked to evaluate their own answers [18], and sampling-based measures such as semantic entropy detect confabulations [10]. ConFTuner trains verbal confidence with a tokenized Brier score [24]. Proximity-informed calibration shows that samples in low-density regions are more over-confident at matched confidence and introduces a matched-confidence bias index [33]; we adopt that index for item popularity. For binary judgments, LLMs are miscalibrated and less reliable on low-frequency entities [23]. Calibration methods for classifiers—temperature scaling [15] and binning [29]—are monotone and therefore cannot reorder a single user’s candidates, a point we make explicit in Section 3.

Yes/no bias and prompt symmetrization. Language models’ binary answers carry prompt- and label-dependent biases. Contextual calibration subtracts the prediction for a content-free input [40], domain-conditional PMI subtracts the prior of the answer surface form [16], and batch calibration estimates the bias from a batch of inputs [41]. Contrast-consistent search scores a statement by combining the probability of the statement and of its negation [6], and crossed symmetrization models framing artifacts that a mirrored prompt cancels [17]. Recent studies find the yes/no bias of LLMs largely global [5, 39]. MIRROR uses the mirrored-question estimator; what is new is the finding that, in personalised recommendation, the cancelled artifact is item-specific and popularity-graded and that cancelling it changes within-user rankings beyond what item-prior subtraction achieves.

Popularity, exposure, and calibration in recommendation. Popularity bias [1] and its causal treatments [32, 37] are long-standing concerns, as are feedback loops that homogenise exposure [7, 27]. Calibrated probabilities for recommendation must account for missing-not-at-random feedback [21], which fully observed data such as KuaiRec [13] makes testable; calibrated probabilities also support per-user list sizes [20]. Data selection for LLM-recommender fine-tuning [26, 28] and denoising of implicit feedback [31] motivate our appendix study of confidence-based pruning.

3 Preliminaries

Yes/no LLM recommender. For user u with history H_u and candidate item i , a prompt $x_q(u, i)$ asks question q (e.g., $q = \text{like}$: “Would this user like the candidate item?”). The recommender’s confidence is the log-odds of the answer token restricted to $\{\text{Yes}, \text{No}\}$:

$$\ell_q(u, i) = \log P(\text{Yes} \mid x_q(u, i)) - \log P(\text{No} \mid x_q(u, i)), \quad \hat{p}_q(u, i) = \sigma(\ell_q(u, i)). \quad (1)$$

Candidates are ranked by ℓ_{like} (equivalently $\hat{p}$).

Labels that match the question. On explicit-feedback data we label a pair $y = 1$ if the user’s rating is ≥ 4 and $y = 0$ if it is ≤ 2 (3-star ratings are dropped), so that $\hat{p}_{\text{like}}$ and y refer to the same event. Calibration against a single held-out next item among many sampled negatives (base rate 1/101) conflates “would like” with “chose next” and is not used for calibration claims.

Why within-user calibration cannot fix ranking.

PROPOSITION 3.1. *Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be strictly increasing and applied to all candidates of a user. Then the within-user ranking, UAUC and every ranking metric are unchanged. If instead $\ell'(u, i) = \ell(u, i) - b(i)$ for an item-dependent offset b , the within-user ranking changes whenever b is not constant on the user’s candidate set.*

The proof is immediate; it explains our earlier negative result that uncertainty-adjusted reranking and popularity-aware post-hoc recalibration did not improve same-candidate ranking, and it identifies the only admissible lever: an *item-dependent* correction.

Decomposition. We posit $\ell_{\text{like}}(u, i) = a(i) + e(u, i) + \epsilon$ and $\ell_{\text{dislike}}(u, i) = a(i) - e(u, i) + \epsilon'$, where $a(i)$ is an item-level acquiescence term (the willingness to answer *Yes* about item i regardless of the question’s polarity) and $e(u, i)$ is personal evidence. Then $a = \frac{1}{2}(\ell_{\text{like}} + \ell_{\text{dislike}})$ and $e = \frac{1}{2}(\ell_{\text{like}} - \ell_{\text{dislike}})$ are identified per pair; the global component of a is rank-inert by Proposition 3.1, so only its *across-item* variation matters.

4 What Does the Yes/No Confidence Measure?

We first characterise token-level $P(\text{Yes})$ of zero-shot and fine-tuned LLM recommenders on rated panels (Table 1), answering Q1–Q5. Each analysis reports a paired user-bootstrap 95% CI.

Q1 (Q1, Q2, Q4): Is the model sure when it is right, and is it wrong because it is unsure? [DATA NEEDED: reliability diagrams of token $P(\text{Yes})$ vs verbalized confidence vs rated labels (ECE, Brier, AUC, UAUC); error anatomy: fraction of correct-but-unsure ($|\hat{p} - 0.5| < \tau$, correct) and confidently-wrong ($|\hat{p} - 0.5| \geq \tau$, wrong) predictions; AUROC of confidence for correctness.]

Q2 (Q5): Are popular items endorsed more confidently—and is it warranted? Following the proximity-bias analysis of Xiong et al. [33], we separate the *level* of confidence from the *calibration gap* at *matched confidence* across item-popularity deciles. **[DATA NEEDED: confidence level by popularity decile; matched-confidence bias index head vs tail; spread of the within-user-centred item term $a(i)$ (SD across items) and Spearman correlation of $a(i)$, valence $v(i)$ and the user-marginalised prior $\pi(i)$ with log-popularity.]**

RQ3: Is the item term familiarity? A pseudonym knockout. We replace every brand name in history and candidate text with a consistent invented pseudonym, and, as a placebo, with another real brand of matched popularity (Appendix A). **[DATA NEEDED: $\Delta\ell$ (real – pseudonym) by popularity decile with head–tail CI; placebo Δ ; change of the matched-confidence bias index; tail UAUC.]**

RQ4 (Q2, Q4 validity): Are “confident errors” errors? Logged negatives may be unexposed positives. On KuaiRec’s fully observed matrix [13] we compare calibration under logged (missing-as-negative) and fully observed labels for the same pairs. **[DATA NEEDED: share of top-decile logged false positives that are true positives; exposure gap ECE(logged)–ECE(full) with CI.]**

Summary. **[DATA NEEDED: one paragraph tying RQ1–RQ4 to the decomposition $\ell = a(i) + e(u, i)$.]**

5 MIRROR: Scoring the Personal Evidence

Mirrored scoring. For each user–candidate pair we ask both questions and score

$$s_{\text{MIRROR}}(u, i) = \ell_{\text{like}}(u, i) - \ell_{\text{dislike}}(u, i) = 2\hat{e}(u, i). \quad (2)$$

Under the decomposition of Section 3 the item term $a(i)$ cancels exactly, so by Proposition 3.1 MIRROR reorders a user’s candidates whenever $a(i)$ varies across them. Both prompts share the history-and-candidate prefix, so with prefix caching the extra cost is one additional single-token decode per pair. A calibrated probability for serving decisions is obtained by temperature-scaling s_{MIRROR} on validation users.

Why not subtract an item prior? Alternatives estimate $a(i)$ from other inputs—a content-free prompt [40], the answer prior [16], the same question without history, or the average over other users’ histories—and subtract it. These estimate the prior of *Yes for item i*, which mixes acquiescence with genuine item quality (an item most users like is not an artifact). The mirrored question isolates the polarity-invariant part only. Whether this distinction matters is an empirical question we test directly (Section 6).

Mirror tuning. Standard yes/no fine-tuning [4] trains only the like question, so nothing constrains the item term. Mirror tuning adds, for each training pair (u, i, y) , the mirrored example with the dislike question and label $1 - y$; the loss is cross-entropy on the single answer token, a proper scoring rule on $P(\text{Yes})$. **[DATA NEEDED: if mirror tuning is retained after M3: one sentence on why it keeps $a(i)$ small, supported by the measured SD of $a(i)$ before/after.]**

Complexity. **[DATA NEEDED: measured throughput: prompts/s for single vs mirrored scoring with prefix caching.]**

6 Experiments

6.1 Setup

Datasets. Rated panels (label ≥ 4 vs ≤ 2): MovieLens-1M and Amazon Reviews 2023 Toys, Beauty and Video Games; users with at least 3 likes and 3 dislikes among their last 20 rated items; user-level train/validation/test split (80/10/10). Next-item panels: the frozen same-candidate protocol of our earlier study (Sports, Toys, Home,

Table 1: Dataset statistics. [DATA NEEDED: fill from panel .meta.json files]

Dataset	Users	Items	Pairs	Like rate
[DATA NEEDED: rows]				

Tools; 10k users; 1 positive + 100 popularity-sampled negatives), on which eight official LLM-enhanced baselines are already evaluated. **[DATA NEEDED: Table 1: statistics.]**

Backbones and regimes. Qwen3-8B and Llama-3.1-8B-Instruct (thinking disabled), zero-shot and LoRA fine-tuned (rank 16, one epoch, three seeds).

Compared methods. (i) Single-prompt confidence: token $P(\text{Yes})$ [4, 38], verbalized probability, self-consistency; (ii) item-prior subtraction: content-free contextual calibration [40], no-history PMI, batch calibration [41], user-swap prior; (iii) trained calibration: temperature scaling [15], Brier fine-tuning [24]; and an equal-compute two-prompt paraphrase ensemble as the placebo for MIRROR.

Metrics. UAUC (primary), AUC, ECE and Brier after validation temperature scaling, tail-item UAUC; NDCG@10/HR@10/MRR on next-item panels; head share of top-10 recommendations. Significance: paired user-bootstrap 95% CI; for fine-tuned models, mean $\pm$ s.d. over 3 seeds and a gain larger than twice the seed s.d.

6.2 Main Results

[DATA NEEDED: Table: UAUC / ECE / tail UAUC for all methods $\times$ 4 datasets $\times$ 2 backbones, zero-shot and fine-tuned; best in bold, CI-significant gains marked.]

6.3 Next-Item Ranking against Official Baselines

[DATA NEEDED: Table: NDCG@10 / HR@10 of raw $P(\text{Yes})$, MIRROR, C-CRP verbalized, and the 8 official baselines on Sports/Toys/Home/Tools; head share of top-10 (Q3, static exposure).]

6.4 Ablations and Analysis

[DATA NEEDED: like-only vs dislike-only vs mirrored; paraphrase set size at equal compute; history length; mirror tuning vs standard SFT; MIRROR on verbalized confidence.]

7 Conclusion

[DATA NEEDED: 3–4 sentences restating the decomposition finding, MIRROR’s gains, and the honest limits.]

Limitations. Same-candidate next-item evaluation with sampled negatives is not full-catalog ranking; rated labels exist only for explicit-feedback domains; the fully observed validity check covers one dataset; pre-training contamination of item text cannot be excluded for public catalogs; all models are 8B-scale. **[DATA NEEDED: any further limitation surfaced by the review loop.]**

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A Pseudonym Knockout Protocol

Brands come from the Amazon store field. Each brand string is replaced, case-insensitively and consistently within an example

(history and candidates), either by a deterministic invented name (hash-seeded syllables) or, as a placebo, by another real brand from the same brand-popularity tercile (fixed derangement). Only candidates whose own title contains their brand string are analysed. Code: `src/confrec/pseudonymize.py`.

B What Did Not Work

[DATA NEEDED: within-user monotone recalibration (rank-inert by Proposition 3.1); popularity-aware post-hoc recalibration; uncertainty-weighted pruning vs matched random pruning over ≥ 5 seeds (Q6); LLM→CF deferral, whose apparent oracle headroom is largely a max-of-two-rankers artifact.]

C Confidence-Based Pruning of Noisy Training Data (Q6)

[DATA NEEDED: detection AUROC of rating-review contradictions by raw entropy, $|\hat{e}|$, evidence; fine-tuning with pruned data vs matched random pruning, ≥ 5 seeds.]

D Reproducibility

Prompts, panel builders, scorers and analysis scripts are released (anonymised link [DATA NEEDED: anonymous.4open.science URL]); all numbers in this paper are produced by the committed scripts.